EXPERIMENT-03

Aim: Understanding Android Components. Create and Android application with multiple activities. Implement the life cycle methods of each activity and observe their execution**.**

Theory:

Steps to Create the Project :

1. Create Project
   - Open Android Studio
   - Click New Project → Empty Views Activity
   - Name it: Practical_3_156
2. Design Layouts
   - activity_main.xml → Add "Next" button
   - activity_second.xml → Add:
     - Username (EditText)
     - Password (EditText)
     - Login button
   - activity_welcome.xml → Add TextView
3. Add Activities
   - Right-click package → New → Activity → Empty Views Activity
   - Create:
     - SecondActivity

○ WelcomeActivity

4. Implement Code
   - MainActivity
     - Use Intent to open SecondActivity
   - SecondActivity
     - Get username & password from EditText
     - Validate inputs (if empty → show Toast)
     - Pass username using putExtra()
   - WelcomeActivity
     - Retrieve username using getStringExtra()
     - Display: "Welcome, [username]!"
5. Check Manifest
   - Ensure all activities are added in AndroidManifest.xml

6. Run App
   - Click Run
   - Test navigation and input

Code:

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/btnNext"
        android:text="Next"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```java
package com.example.experiment_03;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.widget.Button;
import android.content.Intent;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable( $this$enableEdgeToEdge: this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), ( View v,
                                                                              WindowInsetsCompat insets) -> {
            Insets systemBars =
                    insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
                    systemBars.bottom);
            return insets;
        });
        Button btnNext = findViewById(R.id.btnNext);
        btnNext.setOnClickListener( View v -> {
            Intent intent = new Intent( packageContext: MainActivity.this, SecondActivity.class);
            startActivity(intent);
        });
    }
}
```

activity_second.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".SecondActivity">

    <EditText
        android:id="@+id/etUsername"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="text"
        android:hint="User Name"
        tools:layout_editor_absoluteX="7dp"
        tools:layout_editor_absoluteY="22dp" />
    <EditText
        android:id="@+id/etPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:ems="10"
        android:hint="Password"
        android:inputType="textPassword"
        tools:layout_editor_absoluteX="9dp"
        tools:layout_editor_absoluteY="85dp"
        app:layout_constraintTop_toBottomOf="@+id/etUsername"/>
    <Button
        android:id="@+id/btnLogin"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Login"
        tools:layout_editor_absoluteX="56dp"
        tools:layout_editor_absoluteY="139dp"
        app:layout_constraintTop_toBottomOf="@+id/etPassword"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

SecondActivity.java

```java
package com.example.experiment_03;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.content.Intent;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class SecondActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable( $this$enableEdgeToEdge: this);
        setContentView(R.layout.activity_second);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), ( View v,
                                                                              WindowInsetsCompat insets) -> {
            Insets systemBars =
                    insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
                    systemBars.bottom);
            return insets;
        });
```

```java
        EditText etUsername = findViewById(R.id.etUsername);
        EditText etPassword = findViewById(R.id.etPassword);
        Button btnLogin = findViewById(R.id.btnLogin);
        btnLogin.setOnClickListener( View v -> {
            String username = etUsername.getText().toString();
            String password = etPassword.getText().toString();
// Perform login logic here
// For example, check if the username and password are correct
            if (username.isEmpty() || password.isEmpty()) {
// Show an error message if the fields are empty
                Toast.makeText( context: SecondActivity.this,  text: "Please enter both username and password", Toast.LENGTH_SHOR
            } else {
// Login successful, navigate to the next activity
                Intent intent = new Intent( packageContext: SecondActivity.this, WelcomeActivity.class);
                intent.putExtra( name: "username", username);
                startActivity(intent);
            }
        });
    }
}
```

activity_welcome.xml

activity_main.xml | MainActivity.java | activity_second.xml | SecondActivity.java | activity_welcome.xml | Wel

```androidx.constraintlayout.widget.ConstraintLayout
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".WelcomeActivity">

    <TextView
        android:id="@+id/tvWelcome"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="TextView"
        tools:layout_editor_absoluteX="123dp"
        tools:layout_editor_absoluteY="83dp" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

WelcomeActivity.java

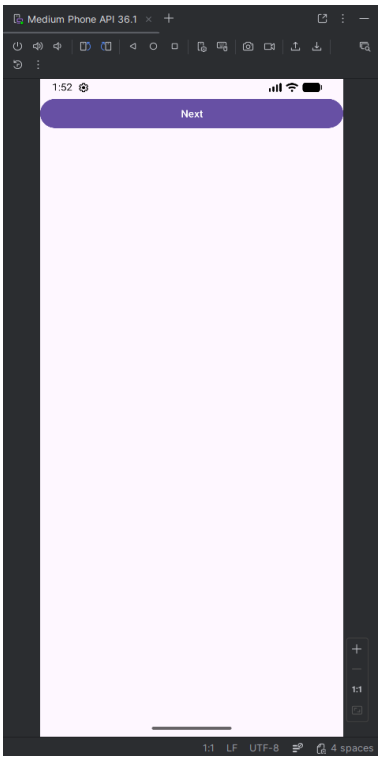
```java
package com.example.experiment_03;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.widget.TextView;


public class WelcomeActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable( $this$enableEdgeToEdge: this);
        setContentView(R.layout.activity_welcome);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), ( View v,
                                                                              WindowInsetsCompat insets) -> {
            Insets systemBars =
                    insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
                    systemBars.bottom);
            return insets;
        });
        TextView tvWelcome = findViewById(R.id.tvWelcome);
//Receive username from the intent
        String username = getIntent().getStringExtra( name: "username");
//Display welcome message with the username
        tvWelcome.setText("Welcome, " + username + "!");
    }
}
```

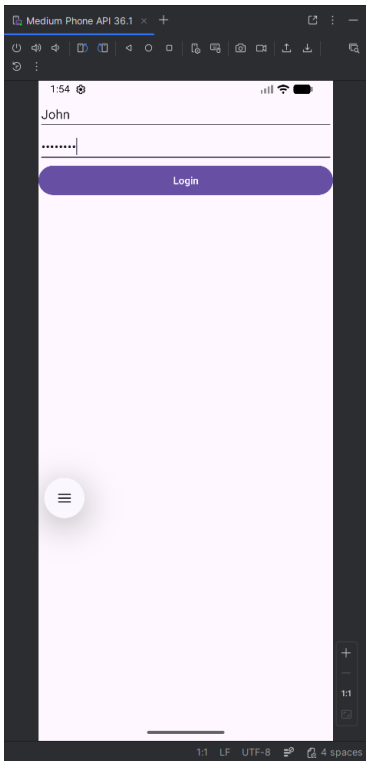
Output:

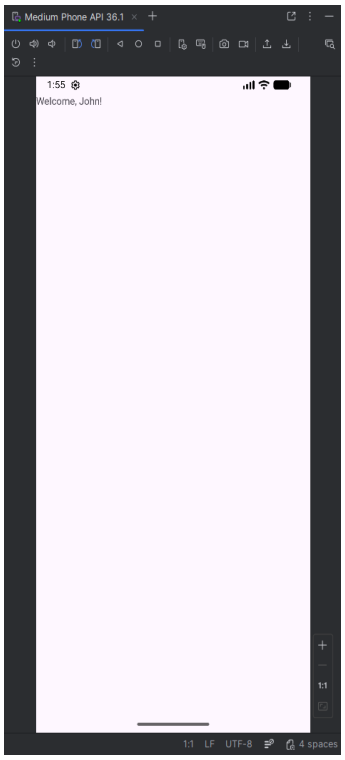
1. Screen Navigation
   - Moves from:
     - Main → Second → Welcome screen using Intent
2. User Input
   - User enters username and password
3. Data Validation
   - If fields are empty → Toast message shown
4. Data Transfer
   - Username passed between activities using:
     - putExtra() and getStringExtra()
5. Dynamic Output

Final screen displays:

Welcome, [username]!







Conclusion:

This practical demonstrates how to build a basic multi-screen Android application using Android Studio. It shows how to navigate between activities using Intents, take user input, validate data, and transfer information between screens. The final output displays a personalized message, making the app interactive and dynamic.